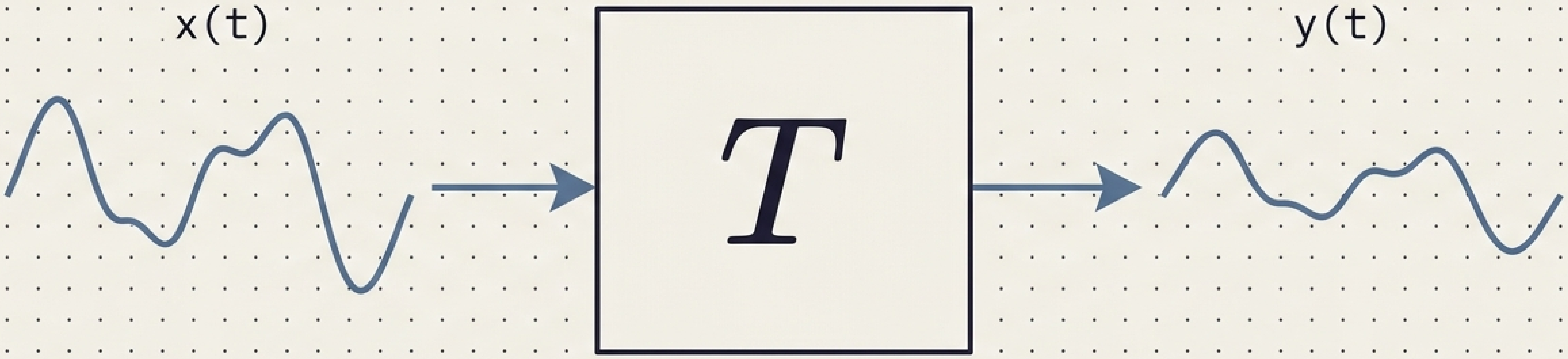
Linear Time-Invariant Systems
The Mathematical Blueprint of Signal Processing

Understanding the mechanics of how mathematical models transform inputs into predictable outputs.

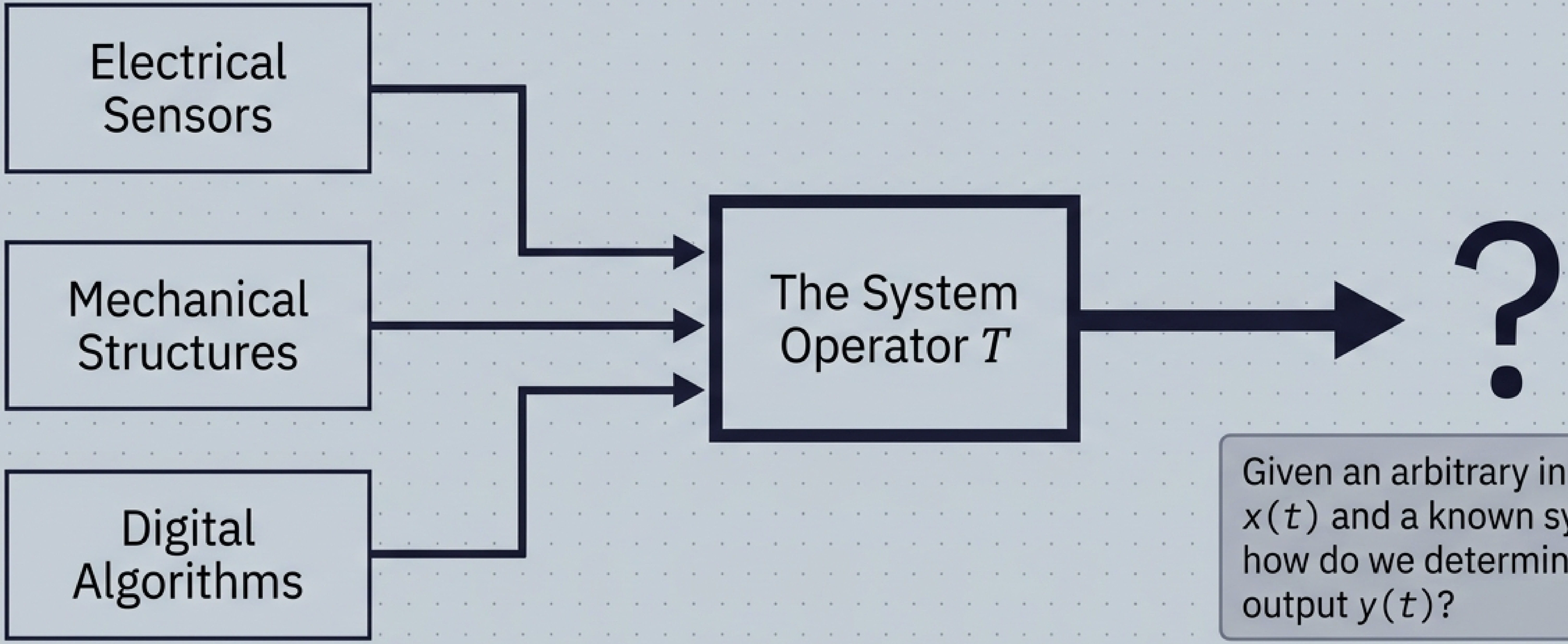


LTI System Transformation

The Core Challenge of System Analysis

A system is a mathematical model that transforms an input signal into an output signal:

$$y(t) = T\{x(t)\}$$



Given an arbitrary input signal $x(t)$ and a known system T, how do we determine the exact output $y(t)$?

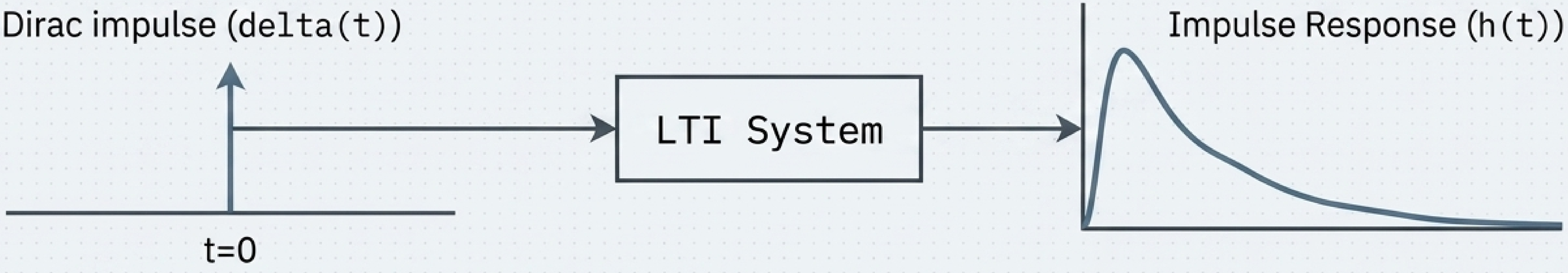
The System Diagnostic Matrix

Property	Mathematical Test	Physical Meaning	Example
Linearity	$T\{ax_1(t) + bx_2(t)\} = aT\{x_1(t)\} + bT\{x_2(t)\}$	Superposition holds. Scaling or summing inputs scales or sums the outputs identically.	$y(t) = 3x(t)$
Time-Invariance	If $x(t) \to y(t)$, then $x(t - t_0) \to y(t - t_0)$	Shifting the input in time causes exactly the same shift in the output.	$y(t) = x(t - 2)$
Causality	$y(t_0)$ depends only on $x(t)$ for $t \leq t_0$	No reliance on the future. Essential for real-time physical systems.	$y(t) = x(t) + x(t - 1)$
Stability (BIBO)	If $\lvert x(t)\rvert \leq M_x < \infty$, then $\lvert y(t)\rvert \leq M_y < \infty$	Bounded inputs yield bounded outputs. The system does not explode.	$y(t) = 2x(t)$
Memory	$y(t_0) = F[x(t_0)]$ vs. reliance on past inputs	Output depends only on current input (Memoryless) vs. past/interval inputs (Dynamic).	$y(t) = \int x(\tau)\, d\tau$

The LTI Superpower: The Impulse Response

If a system is both Linear and Time-Invariant, you unlock a mathematical shortcut.
The system is completely characterized by its response to a single impulse.

Linearity + Time Invariance → LTI System



delta(t) → h(t)

If you know h(t), you know everything about how the system will react to any input in the universe.

The Anatomy of Convolution (Time Domain)

Because the system is LTI, we can slice any input signal into infinitely many weighted, shifted impulses. The total output is simply the sum (integral) of all the individual impulse responses.

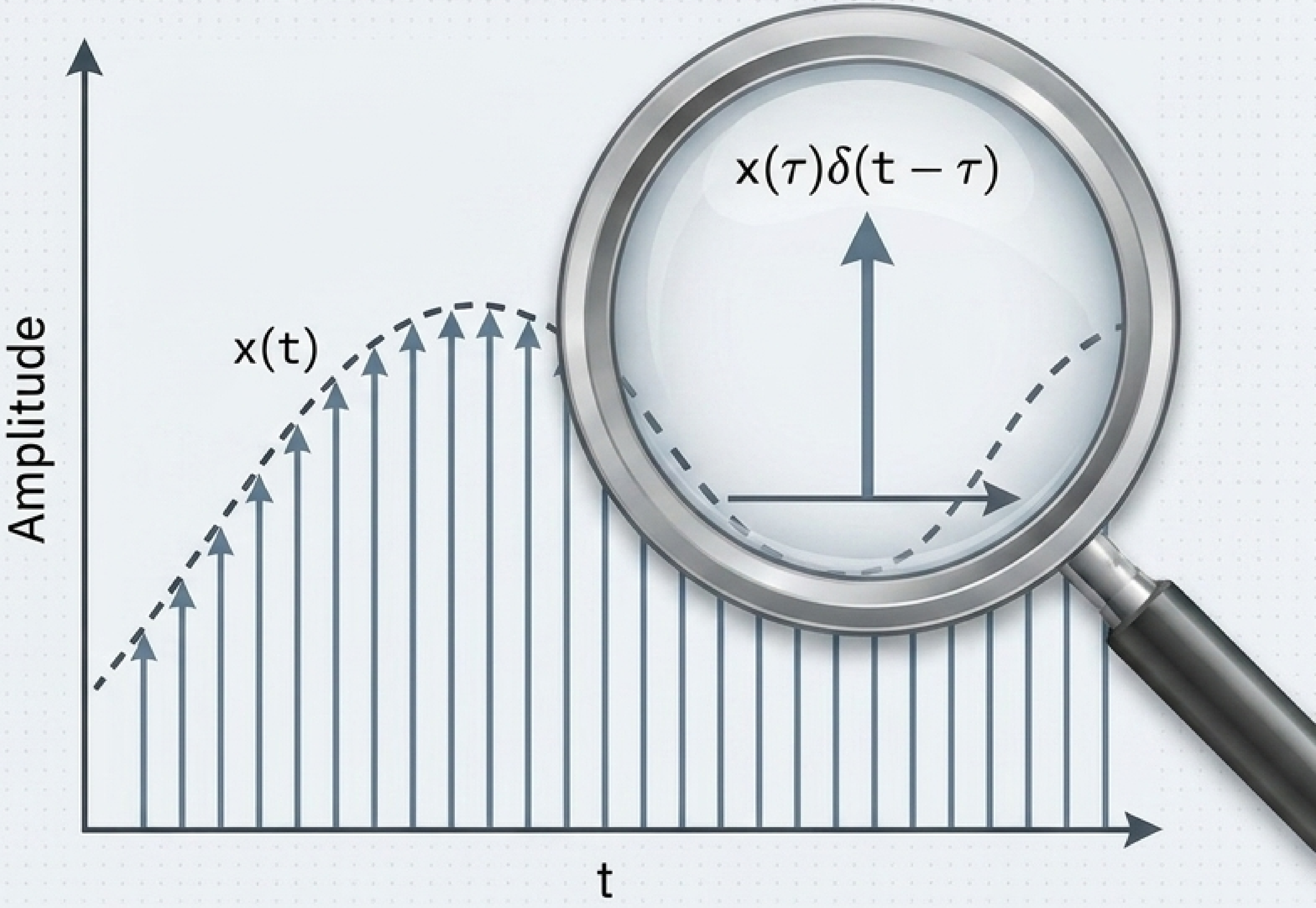
Decompose input x(t) into impulses

↓

Apply system to each impulse → h(t)

↓

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\,d\tau = x(t) * h(t)$$



Graphical Convolution: The Overlap Integral

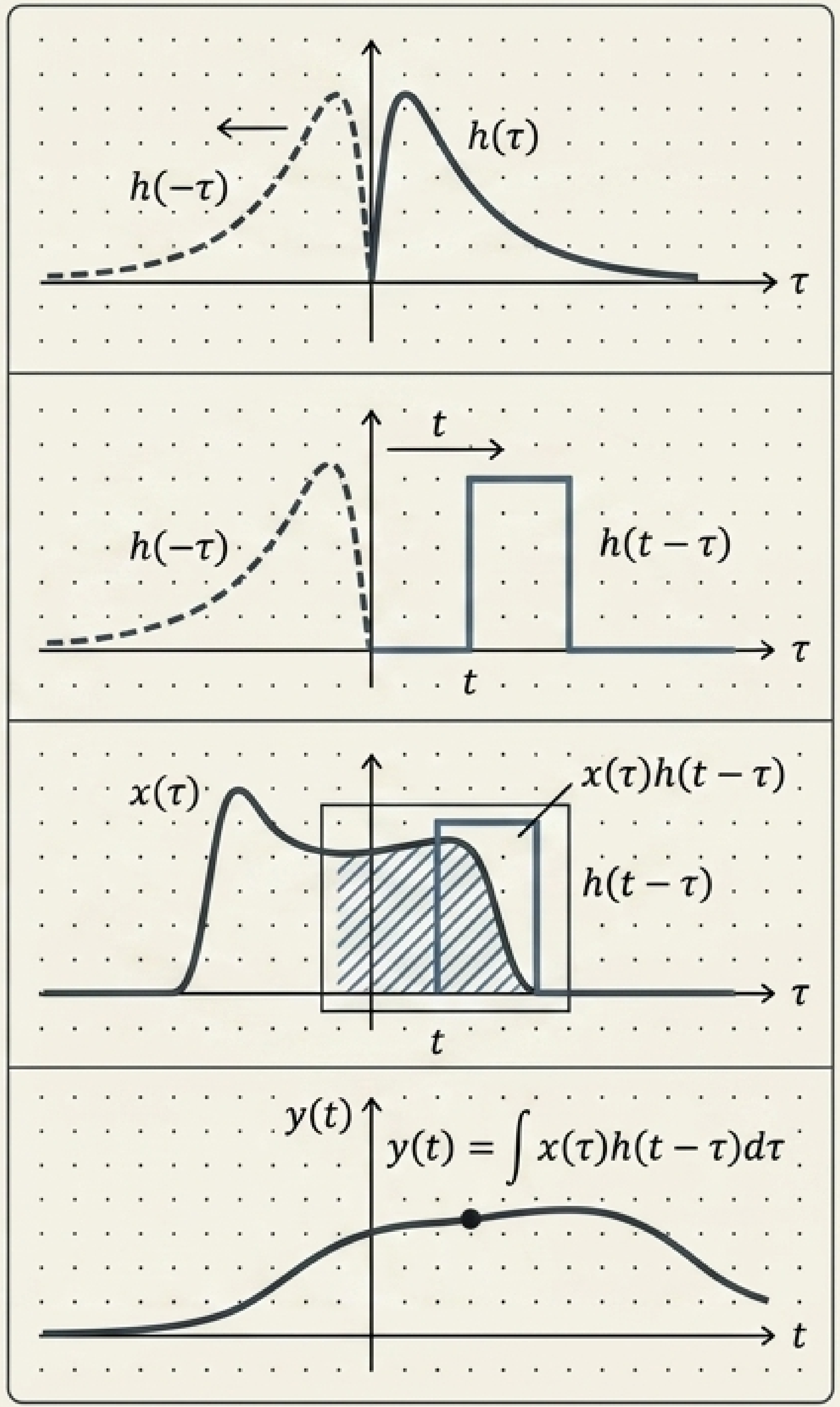
Evaluating $y(t) = x(t) * h(t)$ is a mechanical procedure. Reverse the impulse response, slide it across the input, multiply, and calculate the intersecting area at every moment in time.

1. Reverse

2. Shift

3. Multiply

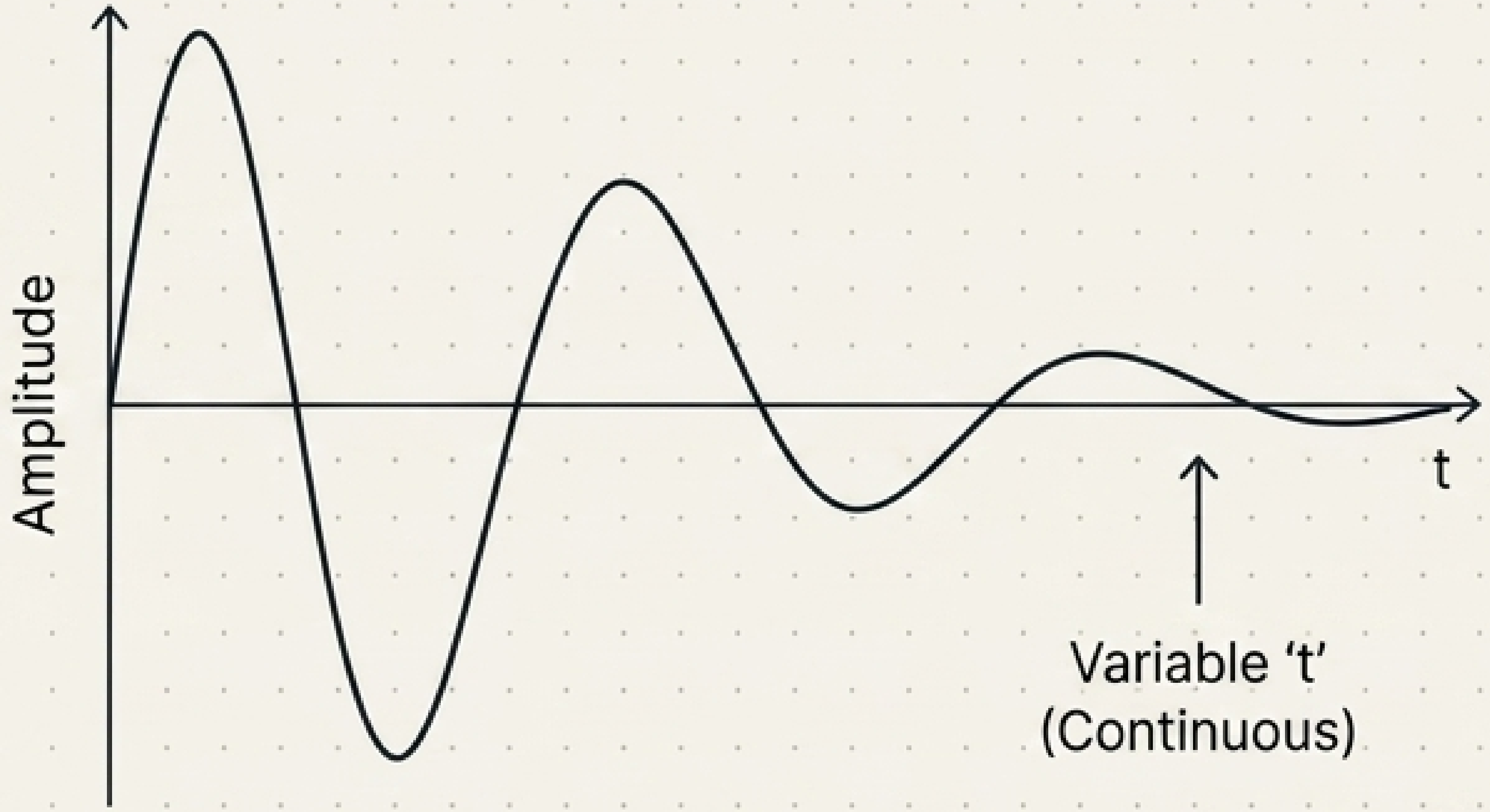
4. Integrate



Universal Logic: Continuous vs. Discrete

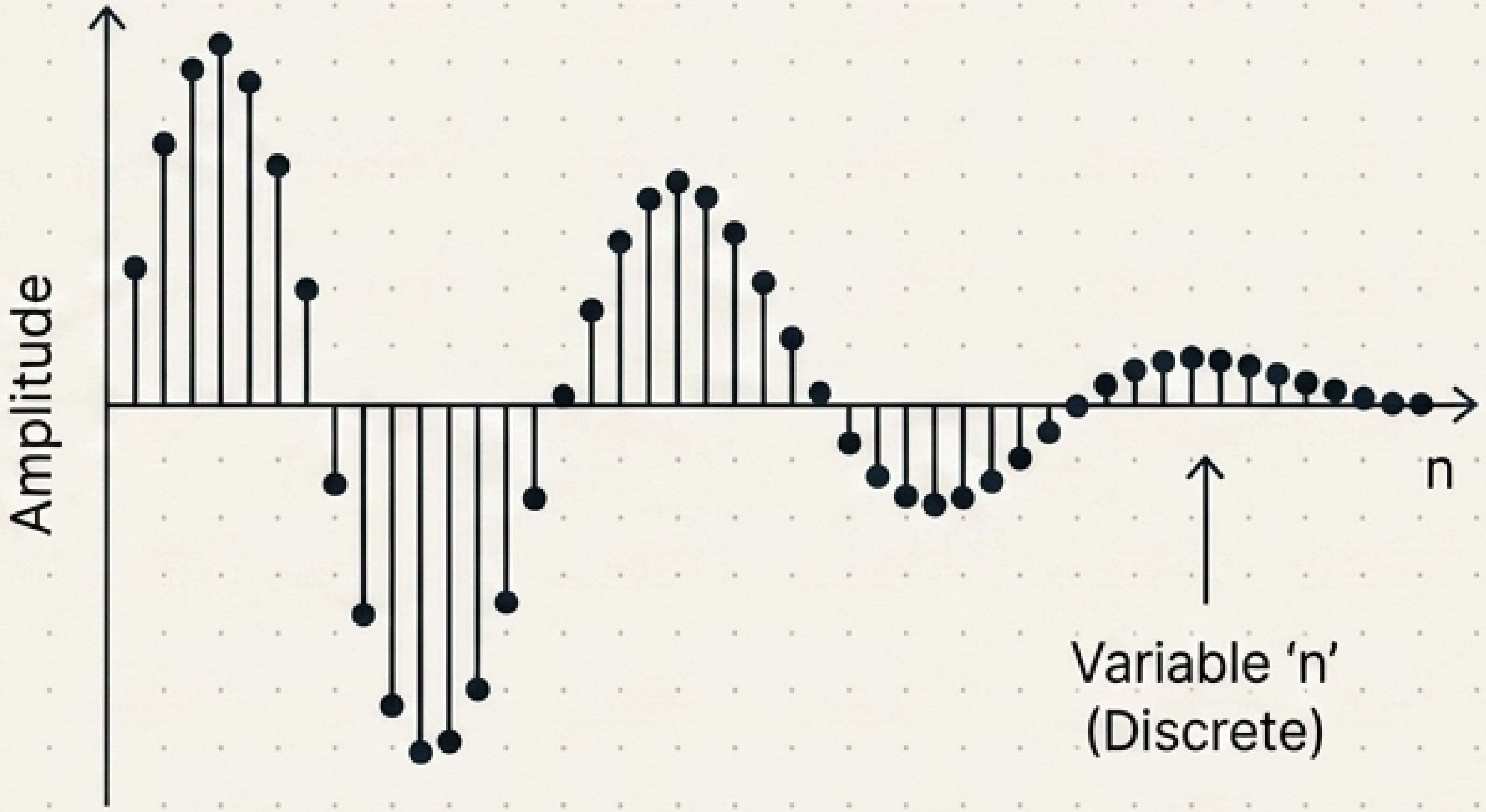
The concepts are identical; the mathematics simply degrade gracefully from continuous integrals to discrete summations.

Continuous Systems



$$y(t) = x(t) * h(t) = \int x(\tau)h(t - \tau)d\tau$$

Discrete Systems



$$y[n] = x[n] * h[n] = \sum x[k]h[n - k]$$

The Fourier Lens (Frequency Domain)

Convolution in the time domain is computationally heavy. The Fourier transform maps signals into the frequency domain, where convolution miraculously simplifies into basic multiplication.

Fourier Prism

$$x(t) * h(t) \longrightarrow \mathcal{F} \longrightarrow X(f) \times H(f)$$

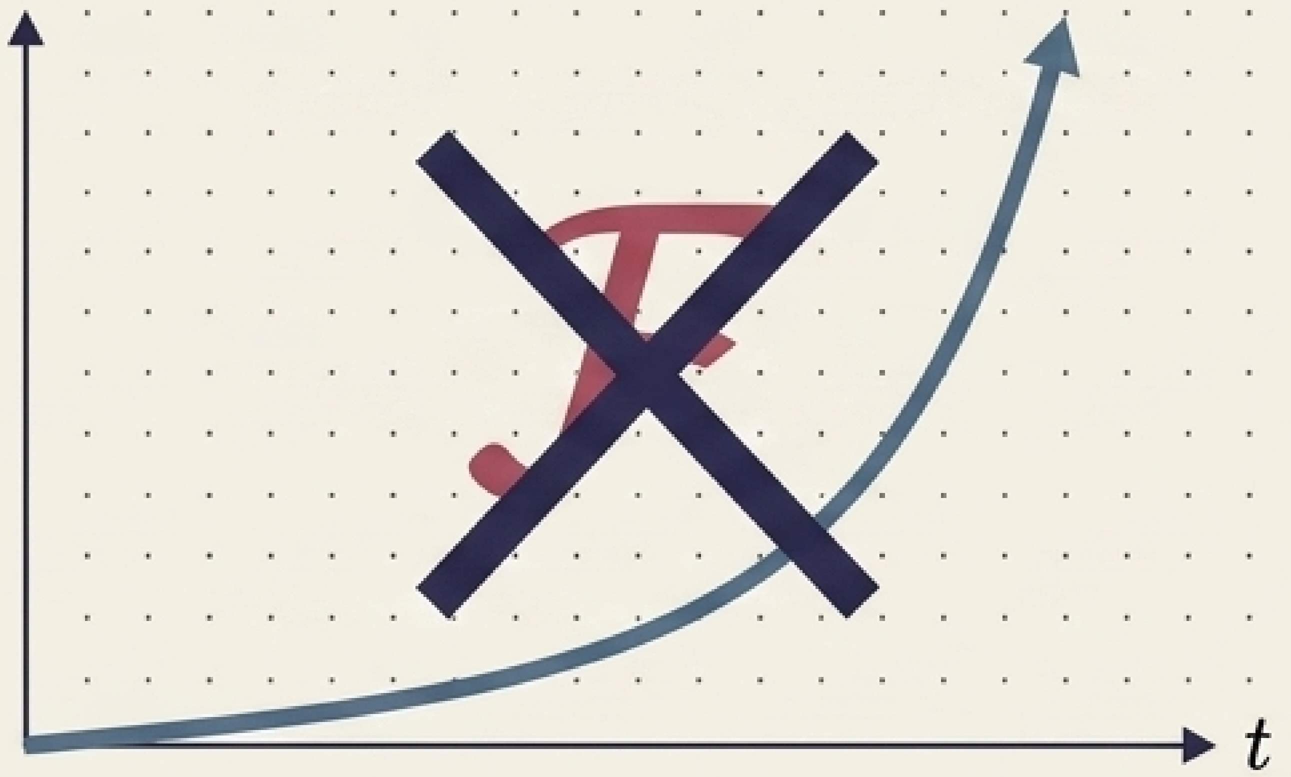
Eigenfunction Insight

Complex exponentials (e^{jwt}) are eigenfunctions of LTI systems. The system keeps the exact frequency, only altering its amplitude and phase. Convolution in time becomes multiplication in frequency.

The Limits of Fourier & The Birth of Laplace

The Fourier transform does not exist for every signal (e.g., growing exponentials). The Laplace transform introduces a complex frequency variable *s* to mathematically tame signals.

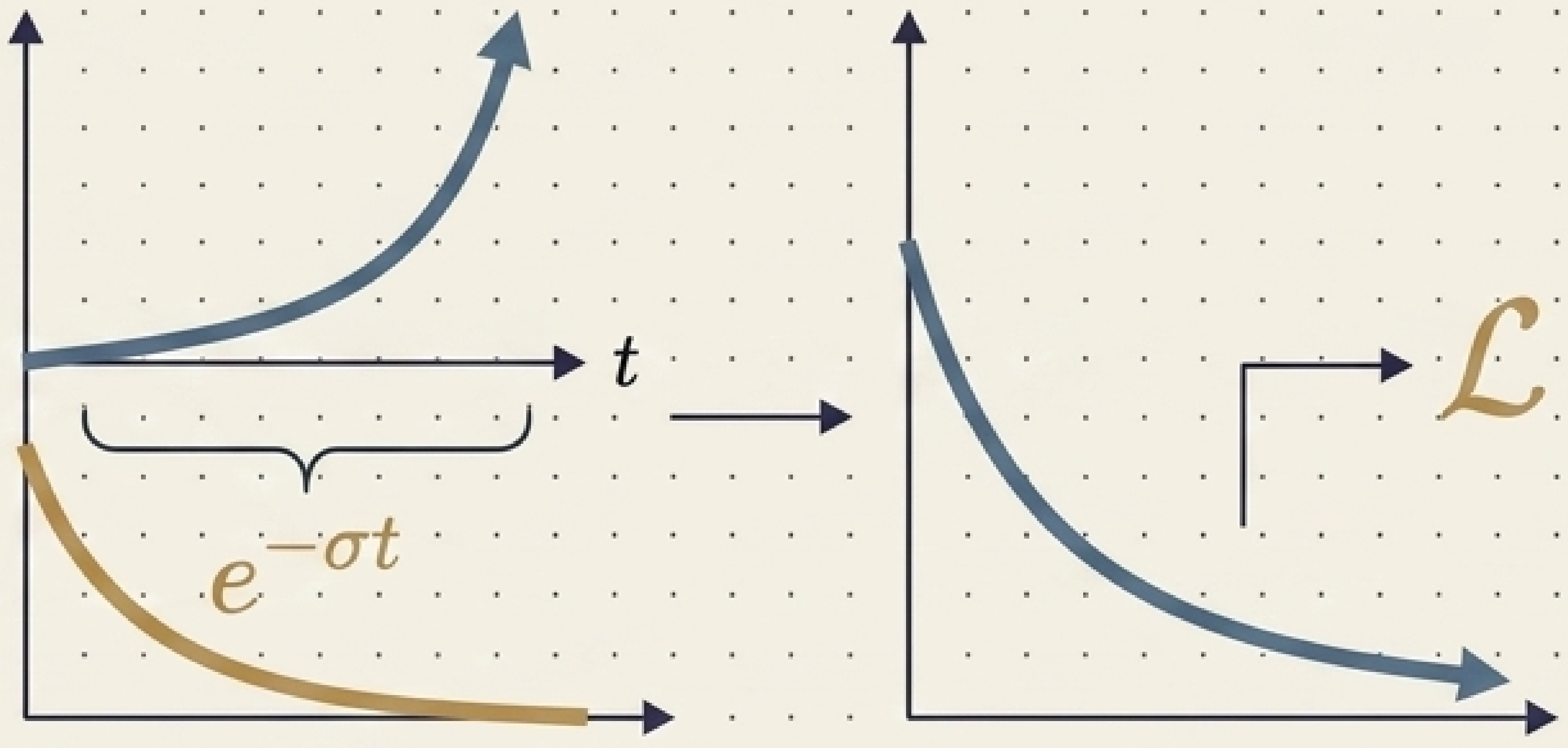
The Limit



Integral fails to converge.

$$\boxed{\int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \to \infty}$$

The Solution

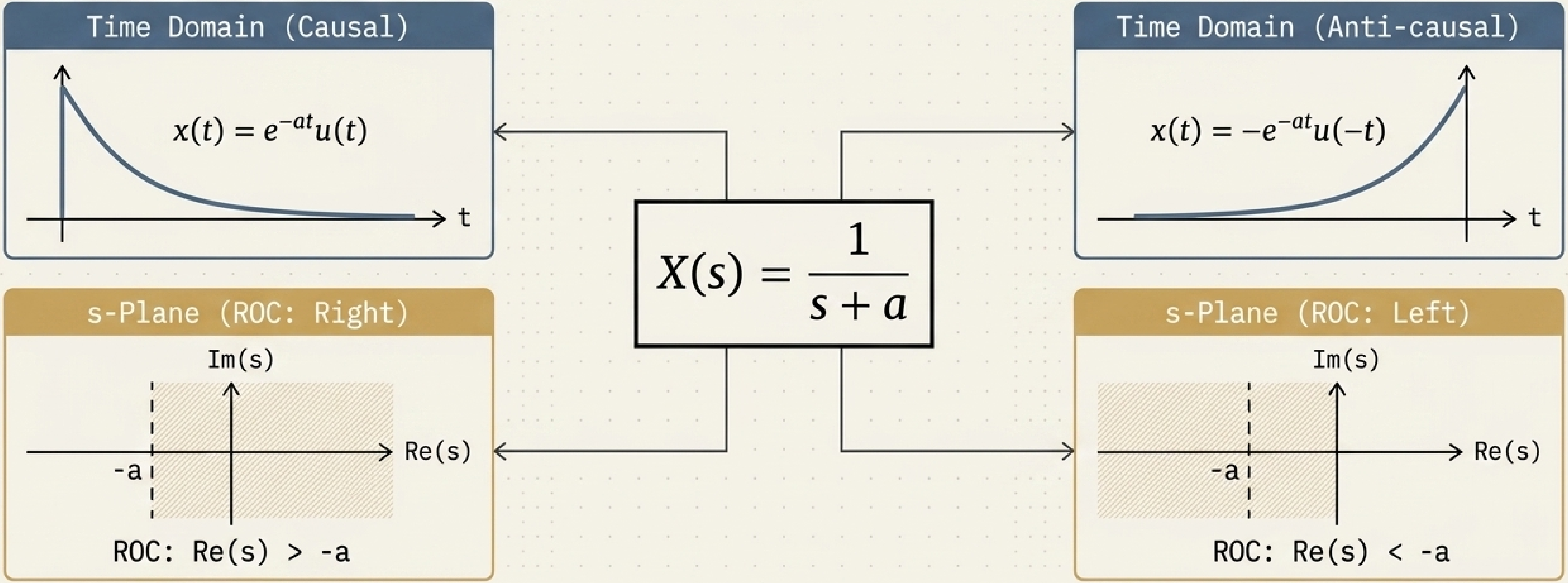


s = sigma + jw. The real part (sigma) forces convergence, while the imaginary part (jw) provides frequency analysis.

$$\boxed{\int_{-\infty}^{\infty} [x(t)e^{-\sigma t}]e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\omega)t}dt}$$

The Territory of the Transform: Region of Convergence (ROC)

Two entirely different signals can have the exact same algebraic Laplace expression.
The transform is incomplete without its territory.



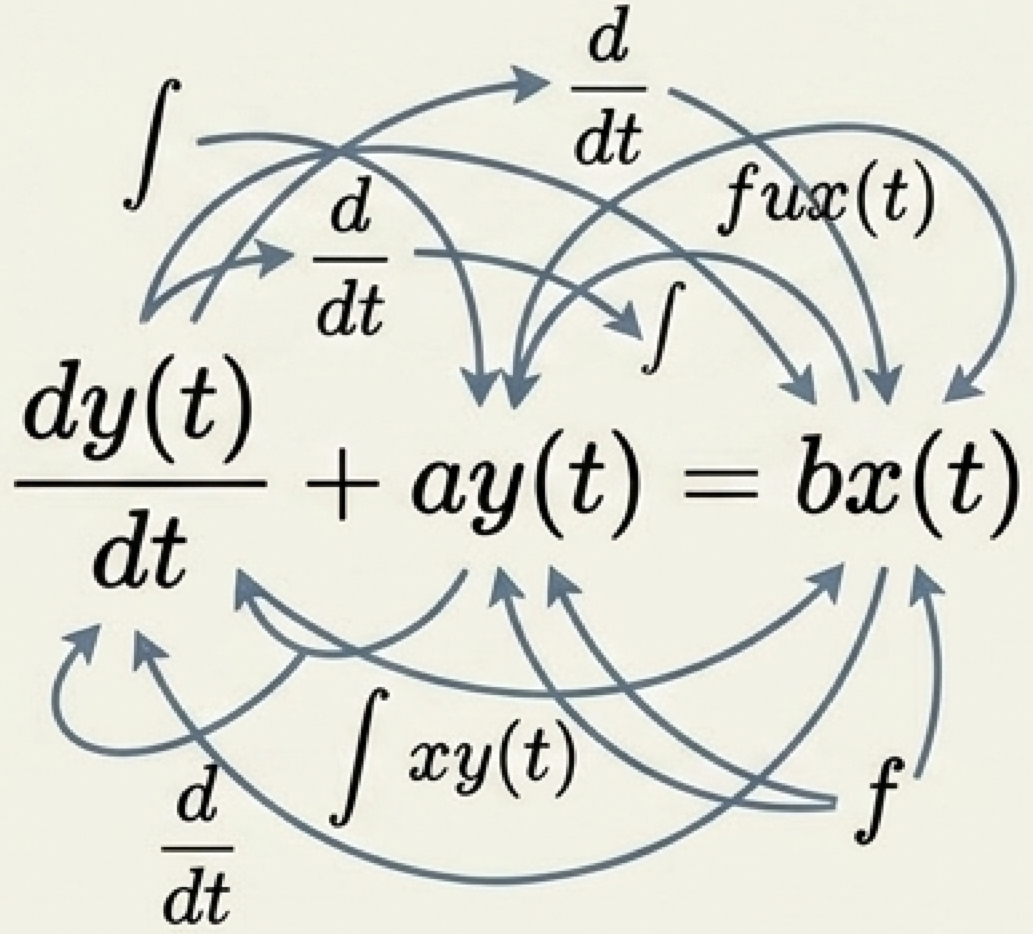
X(s) + ROC = The Complete Signal

The ROC is the specific set of complex values s for which the integral converges.

The Transfer Function: Calculus Becomes Algebra

Laplace transforms derivatives into multiplication by s, reducing complex differential equations to simple algebra.

Differential Equation



Laplace Transform

Inverse Laplace

Algebraic Equation

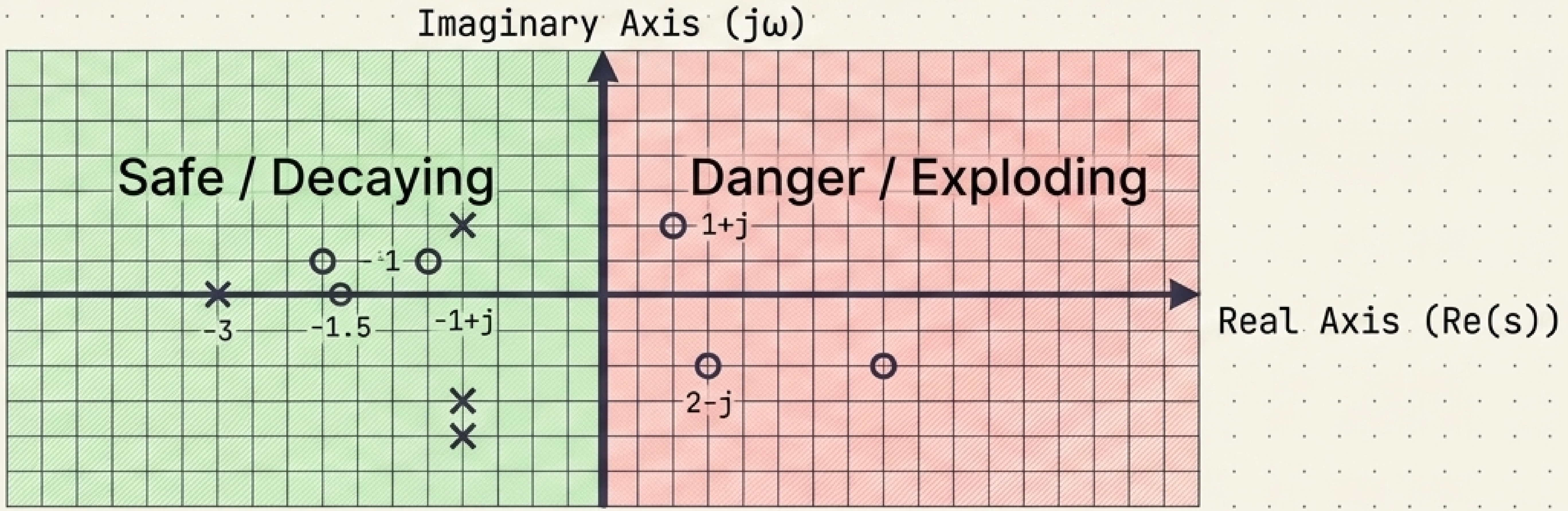
$$(s + a)Y(s) = bX(s)$$

The System Function: H(s) = Y(s) / X(s)

This transfer function provides a compact, complete description of system dynamics, initial conditions, and transients.

Mapping the s-Plane: Poles, Zeros, and Stability

Zeros: Roots where H(s) = 0
Poles: Roots where H(s) approaches infinity (system natural modes)



Stability Law

For a causal rational LTI system, BIBO stability demands that all poles lie strictly in the open left half-plane.

$$\mathrm{Re}(p_i) < 0.$$

SYNTHESIS: The Three Lenses of LTI Systems

Time Domain

$$x(t) \longrightarrow [\ h(t)\ ] \longrightarrow y(t) = x(t) * h(t)$$

Best for understanding waveform shapes, delays, and physical transients.

Fourier Domain

$$X(jw) \longrightarrow [\ H(jw)\ ] \longrightarrow Y(jw) = X(jw)H(jw)$$

Best for bandwidth, filtering, and steady-state frequency.

$$s = jw$$

Laplace Domain

$$X(s) \longrightarrow [\ H(s)\ ] \longrightarrow Y(s) = X(s)H(s)$$

Best for differential equations, poles/zeros, and stability.

The LTI Problem-Solving Workflow

Phase 1: Classify System

(Linear? Causal? Stable? integral $|h(t)|dt < \infty$)

Phase 2: Find Impulse Response $h(t)$

Phase 3: Choose Domain

- Simple Shapes -> Use Time Domain (Convolution)
- Frequency Behavior -> Use Fourier Domain
- Differential Eqs/Transients -> Use Laplace Domain

Common Mistakes to Avoid

1. Assuming $y(t) = x(t) + 5$ is linear (it's not; zero input does not equal zero output).
2. Forgetting time reversal $h(-tau)$ in graphical convolution.
3. Treating $H(s)$ and $H(jw)$ as identical objects.

The Unified LTI Master Framework

Time Domain (Convolution)

$$y(t) = x(t) * h(t)$$
$$x(t) * delta(t) = x(t)$$

Fourier / Laplace Representations

$$Y(f) = X(f)H(f)$$
$$H(s) = Y(s) \,/\, X(s)$$
$$X(jw) = X(s)|_{s=jw}$$

BIBO Stability Conditions

Continuous: $\text{integral } |h(t)|dt < infinity$

Discrete: $\sum_{m}^{m} |h[n]| < infinity$

Poles: $\mathrm{Re}(p_i) < 0$

The Foundational Chain: LTI System → Impulse Response → Convolution → Fourier/Laplace Representation → Total System Control.